

English	(1001CMD303361240009)		Test Pattern
		CLASSROOM CONTACT PROGRAMME (Academic Session : 2024-2025)	NEET (UG) MINOR
			05-05-2024

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E3

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

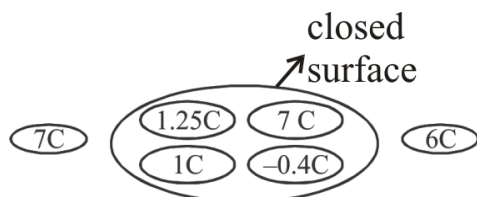
Your Target is to secure Good Rank in Pre-Medical 2025

Telegram: @NEETxNOAH

Topic : CURRENT ELECTRICITY, RAY OPTICS (LENS AND OPTICAL INSTRUMENTS), WAVE OPTICS (TILL LAST LECTURE) [BACK UNIT] : ELECTROSTATICS

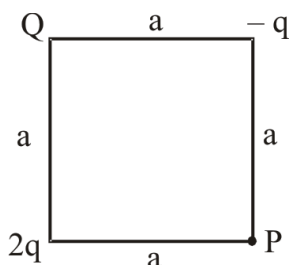
SECTION-A (PHYSICS)

1. What is the electric flux linked with closed surface?



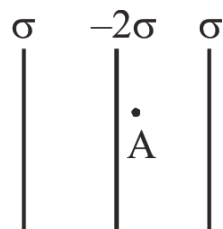
- (1) $10^{11} \text{ N-m}^2/\text{C}$ (2) $10^{12} \text{ N-m}^2/\text{C}$
(3) $10^{10} \text{ N-m}^2/\text{C}$ (4) $8.86 \times 10^{13} \text{ N-m}^2/\text{C}$

2. What is the value of Q so that electric potential at point P becomes zero :-



- (1) $q\sqrt{2}$ (2) $-q\sqrt{2}$ (3) q (4) $-q$

3. Three infinite sheet placed in parallel as shown in figure. Find net electric field at point A



- (1) $\frac{\sigma}{\epsilon_0}$ (2) $\frac{2\sigma}{\epsilon_0}$ (3) $\frac{\sigma}{2\epsilon_0}$ (4) Zero

4.

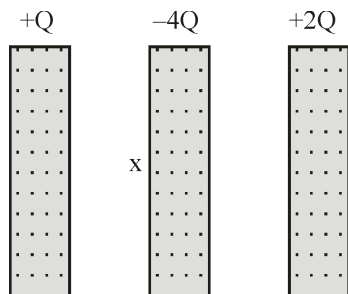


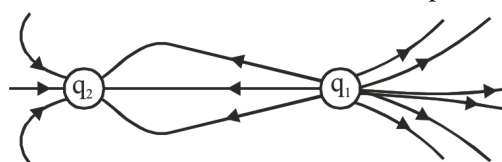
Figure shows three parallel conducting plates placed close to each other. Then charge on surface x is -

- (1) $+\frac{3Q}{2}$ (2) $-\frac{3Q}{2}$ (3) $+\frac{5Q}{2}$ (4) $-\frac{5Q}{2}$

5. Two particle A and B (B is right of A) having charges $27\mu\text{C}$ and $-3\mu\text{C}$ respectively, are held fixed with separation of 20 cm. where should a third charged particle be placed so that it does not experience a net electric force

- (1) 10 cm right of B (2) 10 cm left of A
(3) 5 cm right of B (4) 20 cm left of A

6. For the figure shown, what is the ratio of the charges q_2/q_1 , where the figure shown has a representation of the field lines in the space.



- (1) $-3/2$ (2) $-2/3$ (3) $-4/3$ (4) $-5/2$

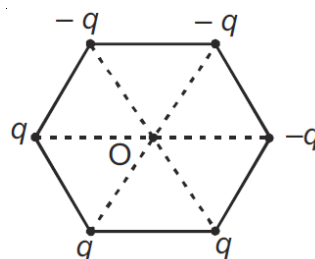
7. Two point charges placed at a distance 'r' in air exert a force 'F'. The distance at which they exert same force when placed in a certain medium (dielectric constant K) is :-

- (1) rK (2) r/K (3) $r/\sqrt{K}$ (4) $r\sqrt{K}$

8. A particle A of mass m and charge Q moves directly towards a fixed particle B, which has charge Q. The speed of A is a V when it is far away from B. The minimum separation between the particles is not proportional to :

- (1) Q^2 (2) $\frac{1}{V^2}$ (3) $\frac{1}{V}$ (4) $\frac{1}{m}$

9. Six point charges are placed at the vertices of a hexagon of side 1m as shown in figure. Net electric field at the centre of the hexagon is

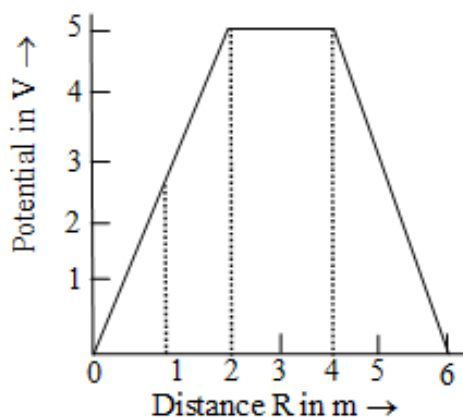


- (1) Zero (2) $\frac{6q}{4\pi\epsilon_0}$ (3) $\frac{q}{\pi\epsilon_0}$ (4) $\frac{q}{4\pi\epsilon_0}$

10. **Assertion** : If a proton and an electron are placed in the same uniform electric field they experience different acceleration

Reason : Electric force on a test charge is independent of its mass.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.
11. In a certain region of space, the potential is given by : $V = k[2x^2 + y^2]$. The electric field at the point (1, 1) has magnitude =
- (1) $k\sqrt{5}$ (2) $2k\sqrt{5}$
 (3) $2k\sqrt{3}$ (4) $4k\sqrt{3}$
12. An electric dipole has the magnitude of its charge as q and its dipole moment is p . It is placed in a uniform electric field E . If its dipole moment is along the direction of the field, the force on it and its potential energy are respectively :-
- (1) $2q.E$ and minimum (2) $q.E$ and $p.E$
 (3) Zero and minimum (4) Zero and maximum
13. The variation of potential with distance R from fixed point is shown in figure. The electric field at $R = 5\text{m}$ is -



- (1) 2.5 V/m (2) -2.5 V/m
 (3) 2/5 V/m (4) -2/5 V/m

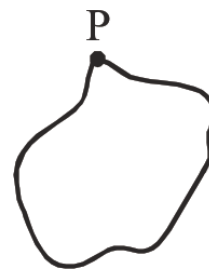
14. A charged particle of charge q and mass m is released from rest in an uniform electric field E . Neglecting the effect of gravity, the kinetic energy of the charged particle after time ' t ' seconds is

- (1) $\frac{Eqm}{t}$ (2) $\frac{E^2 q^2 t^2}{2m}$
 (3) $\frac{2E^2 t^2}{mq}$ (4) $\frac{Eq^2 m}{2t^2}$

15. Two conducting spheres A and B of radius a and b respectively are at same electric potential. The ratio of the surface charge densities of A and B is:-

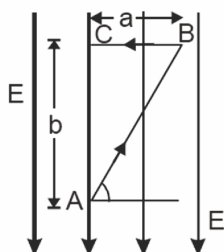
- (1) $\frac{a}{b}$ (2) $\frac{b}{a}$
 (3) $\frac{a^2}{b^2}$ (4) $\frac{b^2}{a^2}$

16. If we have an irregularly shaped conductor as shown and a charge is given to it, choose the correct statement.

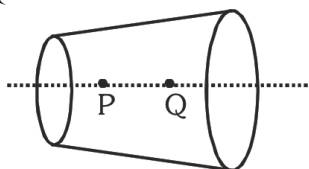


- (1) the electric field and electric potential at P is maximum among all points of conductor.
 (2) the electric field at P is maximum and electric potential is same at all points of conductor.
 (3) the electric field and electric potential at P is same at all points of conductor.
 (4) the electric field and electric potential at P is minimum among all points of conductor.
17. Assume that an electric field $\vec{E} = 30x^2 \hat{i}$ exists in space. Then potential difference $V_A - V_0$, where V_0 is the potential at the origin and V_A the potential at $x = 2\text{m}$ is :-
- (1) 120 V (2) -120 V
 (3) -80 V (4) 80 V

18. The potential difference between points A and B in the given uniform electric field is :



- (1) Ea (2) $E\sqrt{a^2 + b^2}$
 (3) Eb (4) $(Eb/\sqrt{2})$
19. A wire has a non-uniform cross-section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q , then :-



- (1) $v_P = v_Q$ (2) $v_P < v_Q$
 (3) $v_P > v_Q$ (4) data is insufficient
20. The current flowing through a wire depends on time as $I = 3t^2 + 2t + 5$. The charge flowing through the cross-section of the wire in time from $t = 0$ to $t = 2$ sec. is :-
- (1) 22 C (2) 20 C (3) 18 C (4) 5 C
21. There are 8.4×10^{28} free electrons per m^3 in copper. The current in the wire is 0.21 A ($e = 1.6 \times 10^{-19}$ C). Then the drifts velocity of electrons in a copper wire of 1 mm^2 cross section, will be :-
- (1) $2.12 \times 10^{-5} \text{ m/s}$ (2) $0.78 \times 10^{-5} \text{ m/s}$
 (3) $1.56 \times 10^{-5} \text{ m/s}$ (4) $1 \times 10^{-4} \text{ m/s}$
22. Two wires A and B of same material but have length ratio 2 : 3 and radius ratio 1 : 2. If resistance of wire A is 10Ω then what would be resistance of wire B ?
- (1) 2.5Ω (2) 3Ω
 (3) 3.75Ω (4) 4Ω

23. A lamp is placed 6.0 m from a wall. On putting a lens between the lamp and the wall at a distance of 1.2 m from the lamp a real image of the lamp is formed on the wall. The magnification of the image is :-

(1) 3 (2) 4 (3) 5 (4) 6

24. A lens forms a virtual image 4 cm away from it when an object is placed 10 cm away from it. The lens is a lens of focal length :-

(1) Concave, 6.67 cm
 (2) Concave, 2.86 cm
 (3) Convex, 2.86 cm
 (4) May be concave or convex, 6.67 cm

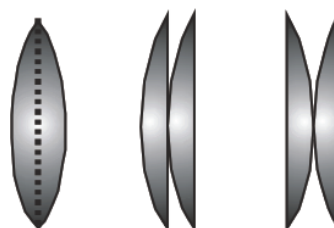
25. A convex lens of focal length 30 cm produces 5 times magnified real image of an object. Object distance from lens will be :-

(1) 36 cm (2) 25 cm (3) 30 cm (4) 150 cm

26. A convex lens of power +6 dioptre is placed in contact with a concave lens of power -4 dioptre. What will be the nature and focal length of this combination ?

(1) Concave, 25 cm (2) Convex, 50 cm
 (3) Concave, 20 cm (4) Convex, 100 cm

27. Two similar plano-convex lenses are combined together in three different ways as shown in the adjoining figure. The ratio of the focal length in three cases will be :-



(1) 2 : 2 : 1 (2) 1 : 1 : 1
 (3) 1 : 2 : 2 (4) 2 : 1 : 1

28. The equations of two interfering waves are $Y_1 = b \cos \omega t$ and $Y_2 = b \cos (\omega t + \phi)$ respectively. Destructive interference will take place at the point of observation for the following value of ϕ :-

(1) 0° (2) 360° (3) 180° (4) 720°

29. It is found that when waves from two coherent sources superpose at a certain point, then the resultant intensity is equal to the intensity of one wave only phase difference between two waves at that point is :-

- (1) zero (2) $\pi/3$
(3) $2\pi/3$ (4) π

30. Intensity of central fringe in interference pattern is 0.01 W/m^2 then find intensity at a point having path difference $\lambda/3$ on screen (in mW/m^2)

- (1) 2.5 (2) 5
(3) 7.5 (4) 10

31. In a simple microscope image is formed at a distance of 50 cm from the lens. If the focal length of lens is 5 cm, then find out its magnifying power.

- (1) 5 (2) 5.5
(3) 0.5 (4) 4.5

32. In astronomical telescope focal length of objective and eye piece is 120 cm and 5 cm respectively. If this telescope is used to view 100 m tall tower 3 km away then find out the angular width of final image formed by eyepiece in normal adjustment.

- (1) $\frac{1}{30}$ radian (2) 0.8°
(3) 30° (4) 0.8 radian

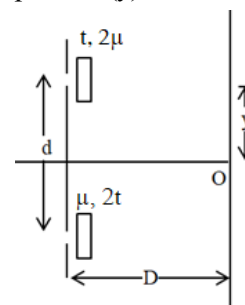
33. In magnifying glass we see the image :

- (1) Virtual, Erect, Large
(2) Virtual, Erect, Small
(3) Virtual, inverted, Large
(4) Real, inverted, Large

34. A telescope has an objective of focal length 100 cm and an eye piece of focal length 5 cm. What is the magnifying power of telescope when final image formed at least distance of distinct vision.

- (1) 20 (2) 24
(3) 120 (4) 40

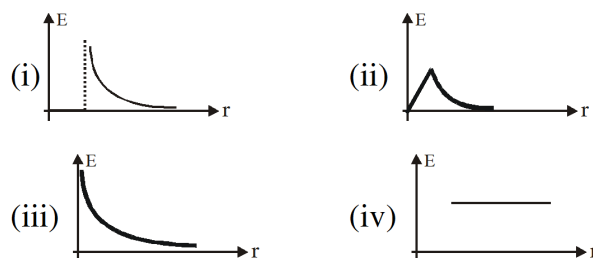
35. In the YDSE shown the two slits are covered with thin sheets having thickness t & $2t$ and refractive index 2μ and μ . Find the position (y) of central maxima.



- (1) zero (2) $\frac{tD}{d}$ (3) $-\frac{tD}{d}$ (4) None

SECTION-B (PHYSICS)

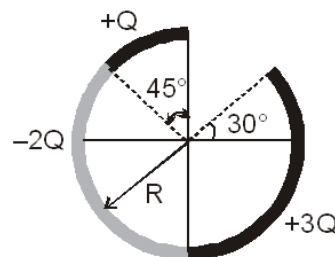
36. Following are graphs of electric field variation with distance (r) for four different charge configurations. Choose the correct correspondence.



- (A) infinitely large charged sheet
(B) infinite line charge
(C) uniformly charged spherical shell
(D) uniformly charged solid sphere

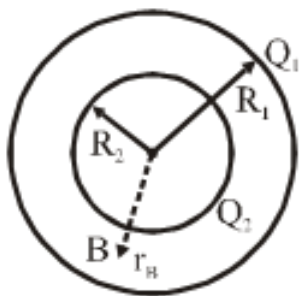
- (1) (i)(B), (ii)(C), (iii)(D), (iv)(A)
(2) (i)(C), (ii)(D), (iii)(A), (iv)(B)
(3) (i)(C), (ii)(D), (iii)(B), (iv)(A)
(4) (i)(D), (ii)(C), (iii)(B), (iv)(A)

37. Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is :



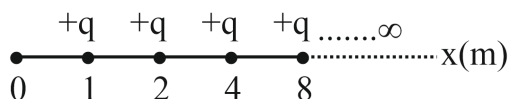
- (1) $\frac{Q}{2\pi\epsilon_0 R}$ (2) $\frac{Q}{4\pi\epsilon_0 R}$ (3) $\frac{2Q}{\pi\epsilon_0 R}$ (4) $\frac{Q}{\pi\epsilon_0 R}$

38. You are given two concentric charged conducting spheres of radii R_1 and R_2 such that $R_1 > R_2$, having charges Q_1 and Q_2 respectively and uniformly distributed over its surface. Calculate E and V at point B whose distance from the centre is r_B as shown in figure :-



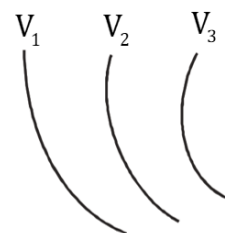
- (1) $E = \frac{Q_1}{4\pi\epsilon_0 r_B^2}, V = \frac{Q_1}{4\pi\epsilon_0 R_1}$
- (2) $E = \frac{Q_2}{4\pi\epsilon_0 r_B^2}, V = \frac{Q_1}{4\pi\epsilon_0 R_1} + \frac{Q_2}{4\pi\epsilon_0 r_B}$
- (3) $E = \frac{(Q_1 + Q_2)}{4\pi\epsilon_0 r_B^2}, V = \frac{Q_1}{4\pi\epsilon_0 R_1} + \frac{Q_2}{4\pi\epsilon_0 r_B}$
- (4) $E = 0, V = 0$
39. Dipole moment of a straight dipole is p . Now it is bent in arc form, this arc makes an angle 60° at its centre. Then new dipole moment will be :-

- (1) p (2) $\frac{p}{\pi}$
- (3) $\frac{2p}{\pi}$ (4) $\frac{3p}{\pi}$
40. Magnitude of electric field at origin due to infinite number of charges as shown in figure is ____.

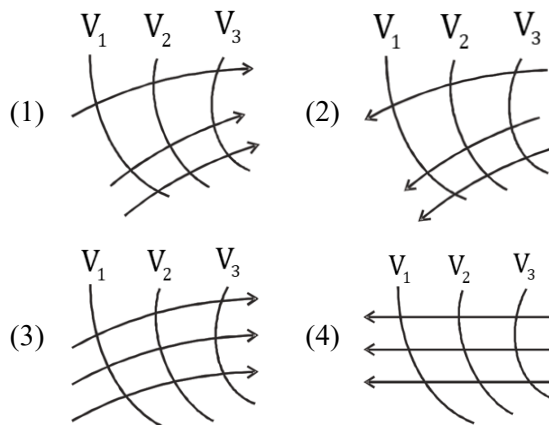


- (1) $\frac{2}{5}Kq$
- (2) $\frac{3}{5}Kq$
- (3) $\frac{4}{5}Kq$
- (4) $\frac{4}{3}Kq$

41. Some equipotential surfaces are shown in the figure such that $V_1 < V_2 < V_3$ in an electric field region.



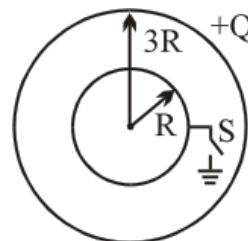
Choose corresponding electric field lines present in this region.



42. An electric dipole is situated in an electric field of uniform intensity E whose dipole moment is p and moment of inertia is I . If the dipole is displaced slightly from the equilibrium position, then the angular frequency of its oscillations is :-

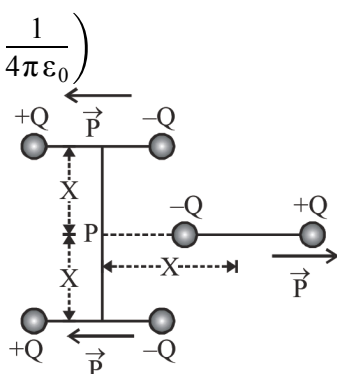
- (1) $\left(\frac{pE}{I}\right)^{1/2}$ (2) $\left(\frac{pE}{I}\right)^{3/2}$
- (3) $\left(\frac{I}{pE}\right)^{1/2}$ (4) $\left(\frac{p}{IE}\right)^{1/2}$

43. Two thin conducting shells of radii R and $3R$ are shown in figure. The outer shell carries a charge $+Q$ and the inner shell is neutral. The inner shell is earthed with the help of switch S . Find the charge attained by the inner shell.



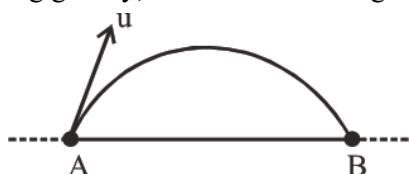
- (1) $\frac{Q}{3}$ (2) $-\frac{Q}{3}$ (3) $\frac{Q}{4}$ (4) $-\frac{Q}{4}$

44. Three identical short dipoles are arranged as shown below. What will be the net electric field at P ($k = \frac{1}{4\pi\epsilon_0}$)



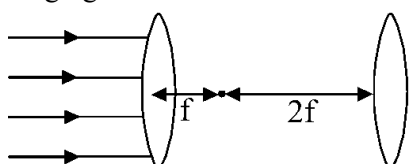
- (1) $\frac{k \cdot p}{x^3}$ (2) $\frac{2kp}{x^3}$ (3) $\frac{4kp}{x^3}$ (4) $\frac{\sqrt{2}kp}{x^3}$

45. In the figure shown there is a large sheet of charge of uniform surface charge density ' σ '. A charge particle of charge ' $-q$ ' and mass ' m ' is projected from a point A on the sheet with a speed ' u ' with angle of projection such that it lands at maximum distance from A on the sheet. Neglecting gravity, find the time of flight :-



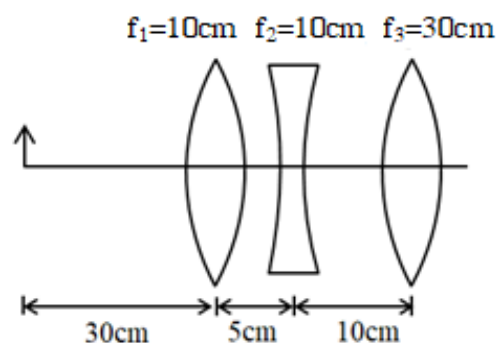
- (1) $\frac{2\sqrt{2}mu\epsilon_0}{\sigma q}$ (2) $\frac{\sqrt{2}mu\epsilon_0}{5\sigma q}$
 (3) $\frac{mu\epsilon_0}{\sqrt{2}\sigma q}$ (4) $\frac{mu\epsilon_0}{2\sqrt{2}\sigma q}$

46. Two convex lenses of focal length f and $2f$ are arranged to have their foci at the same point as shown in the figure. A beam of parallel light of diameter 1 cm is incident into the first lens. The beam emerging from the second lens is:-



- (1) diverging
 (2) converging
 (3) parallel of diameter 0.5 cm
 (4) parallel of diameter 2 cm

47. The position of final image formed by the given lens combination from the third lens will be at a distance of, ($f_1 = +10\text{cm}$, $f_2 = -10\text{cm}$, $f_3 = +30\text{cm}$).



- (1) 15 cm (2) infinity
 (3) 45 cm (4) 30 cm
48. Two identical glass ($\mu_g = 2$) equiconvex lenses of focal length f are kept in contact. The space between the two lenses is filled with water ($\mu_w = 4/3$). The focal length of the combination is :-
- (1) $3f/5$ (2) $f/2$
 (3) $4f/3$ (4) None
49. The intensity of each of the two slits in young's double slit experiment is I_0 . Calculate the minimum separation between the two points on the screen where intensities are $2I_0$ and I_0 (given the fringe width is equal to α) :-
- (1) $\frac{\alpha}{4}$ (2) $\frac{\alpha}{3}$
 (3) $\frac{\alpha}{12}$ (4) None of these
50. In a compound microscope focal length of objective and eye piece are 2 cm and 3 cm respectively. If distance between both the lenses are 19 cm then find out magnifying power of compound microscope when final image formed at infinity.

- (1) $\frac{475}{6}$ (2) $\frac{175}{3}$
 (3) $\frac{225}{3}$ (4) None

Topic : SOLUTION, ELECTROCHEMISTRY, OPTICAL ISOMERISM, CONFORMATIONAL ISOMERISM, FUNDAMENTAL CONCEPTS IN ORGANIC REACTIONS MECHANISM (UPTO CLASS) [BACK UNIT]
: CHEMICAL KINETICS

SECTION-A (CHEMISTRY)

51. The value of Henry's constant for some gases at 293 K is given below. Arrange the gases in the increasing order of their solubility.

He : 144.97 kbar, H₂ : 69.16 kbar, N₂ : 76.48 kbar,
O₂ : 34.86 kbar

- (1) He < N₂ < H₂ < O₂
- (2) O₂ < H₂ < N₂ < He
- (3) H₂ < N₂ < O₂ < He
- (4) He < O₂ < N₂ < H₂

52. Decimolar solution of potassium ferricyanide, K₃[Fe(CN)₆] has osmotic pressure of 3.94 atm at 27°C. Hence approximate percentage ionisation of the solute is

- (1) 10% (2) 20%
- (3) 30% (4) 40%

53. A 5% (w/V) solution of cane sugar is isotonic with 1% (w/V) solution of a non volatile solute. The molecular weight of non volatile solute is (Molecular weight of cane sugar = 342 g) :-

- (1) 34.2 (2) 171.2
- (3) 68.4 (4) 136.8

54. **Assertion :** Pressure does not have any effect on solubility of solids in liquids.

Reason : Solids and liquids are highly incompressible.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

55. **Assertion :-** Molar conductivity of an electrolytic solution increases on dilution.

Reason :- Decrease in conductivity is caused by increase in volume.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false.
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

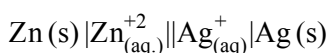
56.

	Column-I		Column-II
(a)	Kohlrausch law	(p)	$\frac{\Lambda_m^c}{\Lambda_m^o}$
(b)	Molar conductance (Λ_m)	(q)	$\frac{1}{R} \times \frac{l}{A}$
(c)	Specific conductance $\rightarrow (\kappa)$	(r)	$\Lambda_m^o \text{ of } \text{Ca}_3(\text{PO}_4)_2 = 3\lambda_+^0 + 2\lambda_-^0$
(d)	Degree of ionization of weak electrolyte	(s)	$\frac{\kappa \times 1000}{M}$

Which of the following option show correct matches

- (1) (a – r), (b – p), (c – q), (d – s)
- (2) (a – s), (b – p), (c – q), (d – r)
- (3) (a – r), (b – s), (c – q), (d – p)
- (4) (a – p), (b – s), (c – q), (d – r)

57. Calculate E°_{cell} for following cell



$$E^{\circ}_{\text{Zn}^{+2}/\text{Zn}} = -0.76\text{V}; E^{\circ}_{\text{Ag}^{+}/\text{Ag}} = 0.80\text{V}$$

- (1) 0.04 V (2) –1.56 V
- (3) 1.56 V (4) 0.84 V

58. **Assertion (A)** :- According to collision theory product can form only when the required orientation and energy conditions are met.

Reason (R) :- All collisions between reactants yield the desired product.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is incorrect but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

59. **Assertion (A)** : Rate of reaction increases when temperature is increased.

Reason (R) : Increasing the temperature increases the fraction of molecules which collide with energy greater than E_a .

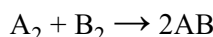
- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) Both A and R are false.

60. Match the column-I & column-II :-

	Column I (Character of reaction)		Column II (Order)
(A)	Reaction with identical rate and rate constant	(p)	Zero
(B)	Reaction rate doubles on increasing concentration four times	(q)	First
(C)	Half-life period is inversely proportional to initial concentration	(r)	Second
(D)	Half-life period is independent of initial concentration	(s)	Half

- (1) (A) $\rightarrow$ (p); (B) $\rightarrow$ (s); (C) $\rightarrow$ (r); (D) $\rightarrow$ (q)
- (2) (A) $\rightarrow$ (p); (B) $\rightarrow$ (r); (C) $\rightarrow$ (s); (D) $\rightarrow$ (q)
- (3) (A) $\rightarrow$ (p); (B) $\rightarrow$ (q); (C) $\rightarrow$ (r); (D) $\rightarrow$ (s)
- (4) (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q)

61. Hypothetical reaction,



is completed using given mechanism.



Order of complete reaction is -

- (1) 2 (2) 1 (3) $1\frac{1}{2}$ (4) 0

62. Which of the following is the example of maximum Boiling azeotropic solution ?

- (1) 20.2% HCl + 79.8 % H₂O
- (2) 32% HNO₃ + 68% H₂O
- (3) 95.5% C₂H₅OH + 4.5 % H₂O
- (4) 79.8% HCl + 20.2 % H₂O

63. Which of the following shows maximum molar conductivity ?

- (1) 0.1 M NaCl (2) 0.01 M NaCl
- (3) 0.5 M NaCl (4) 0.001 M NaCl

64. Which one of the following statement is wrong :-

- (1) Conductivity increases on dilution.
- (2) Conductivity decreases on dilution.
- (3) Equivalent conductivity increases on dilution
- (4) Molar conductivity increases on dilution

65. Which of the following are correct

(A) Λ_m^0 of weak electrolyte can be obtained by extrapolation of Λ_m to Zero concentration

(B) $\Lambda_m^0(\text{KCl}) - \Lambda_m^0(\text{NaCl}) = \Lambda_m^0(\text{KBr}) -$

$\Lambda_m^0(\text{NaBr}) = \Lambda_m^0(\text{KI}) - \Lambda_m^0(\text{NaI})$

(C) Concentration of weak acid

$$(C) = \frac{\Lambda_m^0}{\Lambda_m^2} \cdot K_a (\Lambda_m^0 - \Lambda_m)$$

(D) Λ_{eq}^0 of $\text{Al}_2(\text{SO}_4)_3 = 2\lambda_{\text{eq}}^0(\text{Al}^{+3}) + 3\lambda_{\text{eq}}^0(\text{SO}_4^{-2})$

- (1) A,B,C,D (2) B,C,D
- (3) A,B,C (4) B,C

66. **Statement-I** : For measuring the resistance of an ionic solution, D.C source of power is used.

Statement-II : Passage of direct current through ionic solution over a long period can lead to change in its composition.

- (1) Statement-I is true, Statement-II is true and Statement-II is correct explanation for Statement-I
 (2) Statement-I is true, Statement-II is true but Statement-II is NOT the correct explanation for Statement-I
 (3) Statement-I is true but Statement-II is false
 (4) Statement-I is false but Statement-II is true

67. Λ_m^0 for NaCl, HCl and NaOH are 126.4, 425.9, and 249.2 $\text{Scm}^2 \text{mol}^{-1}$ respectively calculate Λ_m^0 value of water ?

- (1) 274.35 $\text{Scm}^2 \text{mol}^{-1}$
 (2) 303.1 $\text{Scm}^2 \text{mol}^{-1}$
 (3) 606.2 $\text{Scm}^2 \text{mol}^{-1}$
 (4) 548.7 $\text{Scm}^2 \text{mol}^{-1}$

68. The resistance of a cell is 250 ohm which is filled with 0.02 M CaI_2 solution having cell constant 0.9 cm^{-1} then calculate its molar conductivity ?

- (1) 90 $\text{Scm}^2 \text{mol}^{-1}$
 (2) 180 $\text{Scm}^2 \text{mol}^{-1}$
 (3) 30 $\text{Scm}^2 \text{eq}^{-1}$
 (4) 120 $\text{Scm}^2 \text{mol}^{-1}$

69. The rate constant for the reaction ; $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$ is $5 \times 10^{-3} \text{ sec}^{-1}$. If the rate of reaction is $2.5 \times 10^{-3} \text{ M sec}^{-1}$, then the concentration of N_2O_5 is :

- (1) 1 M (2) 2 M
 (3) 0.5 M (4) 5 M

70. The half life period and initial concentration for a reaction are as follows. What is order of reaction ?

Initial concentration	350	540	158
$t_{1/2}$ (hr)	425	275	941

- (1) Zero (2) 1 (3) 2 (4) 3

71. A first order reaction has half life 12 hours then what percentage of the reactant will remain after 24 hours ?

- (1) 75% of initial (2) 87.5% of initial
 (3) 25% of initial (4) 93.75% of initial

72. A zero order reaction has $k = 0.025 \text{ Ms}^{-1}$ for the disappearance of A. What will be concentration of A after 15 sec. if the initial concentration of A is 0.6 M ?

- (1) 0.06 M (2) 0.225 M
 (3) 0.125 M (4) 0.5 M

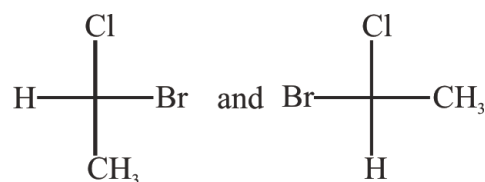
73. In the reaction $2\text{SO}_3(\text{g}) \rightarrow 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ the rate of appearance of O_2 is 0.016 kg.hr^{-1} . What is the rate of formation of SO_2 in kg.hr^{-1} ?

- (1) 0.016 kg.hr^{-1} (2) 0.001 kg.hr^{-1}
 (3) 0.032 kg.hr^{-1} (4) 0.064 kg.hr^{-1}

74. Which conformer of cyclohexane is most stable ?

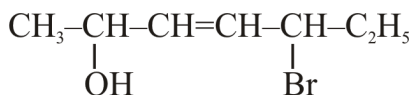
- (1) Chair form (2) Twist boat
 (3) Boat form (4) Half chair

75. What is the relationship between the structures shown below?



- (1) enantiomers
 (2) diastereomers
 (3) configurational isomers
 (4) identical compounds

76. How many total stereo isomers are possible of this structure :-



- (1) 2 (2) 4 (3) 8 (4) 16

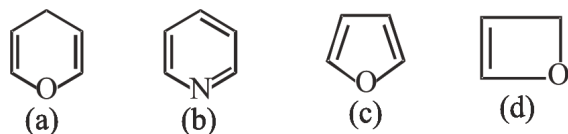
77. Correct order of $-I$ effect is :-

- (1) $-\text{F} > -\text{Cl} > -\text{Br}$
 (2) $-\text{NO}_2 > -\text{F} > -\text{OH}$
 (3) $-\text{OR} > -\text{OH} > -\text{NH}_2$
 (4) All are correct

78. Which Resonating structure is least stable ?

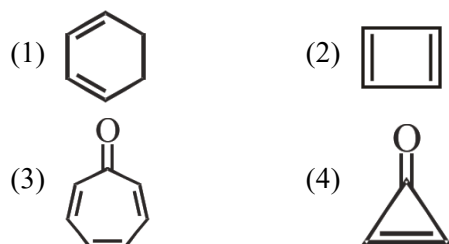
- (1) $\text{CH}_2 = \text{CH} - \text{NH}_2$
 (2) $\ominus \text{CH}_2 - \text{CH} = \text{NH}_2^+$
 (3) $\ominus \text{CH}_2 - \text{CH}^+ - \text{NH}_2$
 (4) Cannot be determined

79. Which of the following is aromatic :-



- (1) a and b (2) b and c
 (3) c and d (4) a and d

80. Which of the following molecules is unstable at room temperature?



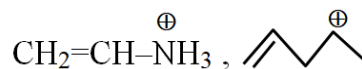
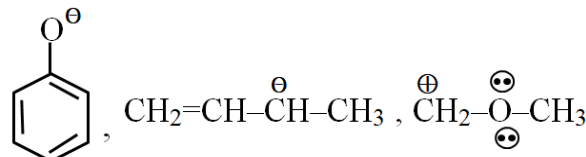
81. Select correct statement ?

- (1) Carbocation has complete octet.
 (2) Carbon free radical is diamagnetic
 (3) Triplet carbene has vacant orbital
 (4) Singlet carbene has vacant orbital

82. Incorrect statement about inductive effect is :

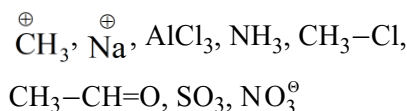
- (1) I -effect is complete shifting of π -electron
 (2) I -effect of H is consider as zero.
 (3) $-\text{NH}_3^+$ group show $-I$ effect
 (4) $-\text{OH}$ group show $-I$ effect.

83. How many of following show resonance.



- (1) 5 (2) 4
 (3) 3 (4) 2

84. How many of following are electrophile



- (1) 8 (2) 3
 (3) 5 (4) 2

85. Select incorrect statement about nucleophile.

- (1) Nu^- are e^- rich and give e^- pair to reactive site of substrate.
 (2) Lewis bases are Nu^-
 (3) Lewis acids are not nucleophile
 (4) H_3O^+ is nucleophile

SECTION-B (CHEMISTRY)

86. Which solution will have the highest boiling point :-

- (1) 1% glucose in water
 (2) 1% sucrose in water
 (3) 1% NaCl in water
 (4) 1% CaCl_2 in water

87. Match the following column-I and column-II on the basis of data given below

$$E_{\text{Li}^+/\text{Li}}^\circ = -3.5\text{V}, E_{\text{Au}^{3+}/\text{Au}}^\circ = 1.40\text{V}$$

Column-I		Column-II	
(A)	Li	(I)	Ion which is the strongest oxidising agent
(B)	Au	(II)	Ion which is the weakest oxidising agent
(C)	Au^{3+}	(III)	Metal which is strongest reducing agent
(D)	Li^+	(IV)	Least reactive metal

- (1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
 (2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
 (3) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
 (4) (A)-(IV), (B)-(III), (C)-(II), (D)-(IV)

88. Using the data given below find out the strongest reducing agent :

$$E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}}^\circ = 1.33\text{V}; E_{\text{Cl}_2/\text{Cl}^-}^\circ = 1.36\text{V}$$

$$E_{\text{MnO}_4^{2-}/\text{Mn}^{2+}}^\circ = 1.51\text{V}; E_{\text{Cr}^{3+}/\text{Cr}}^\circ = -0.74\text{V}$$

- (1) Cl^-
 (2) Cr
 (3) $\text{Cr}_2\text{O}_7^{2-}$
 (4) Mn

89. For a chemical reaction temperature changes from 10°C to 40°C then rate of reaction will be

(Assume $\mu = 2$)

- (1) 8 times of initial
 (2) 6 times of initial
 (3) 4 times of initial
 (4) 2 times of initial

90. **Assertion (A)** : An elementary reaction cannot have fractional order.

Reason (R) : Stoichiometric coefficients in an elementary reaction can be fractional.

- (1) Both assertion and reason are correct and reason is a correct explanation of assertion.
 (2) Both assertion and reason are correct but reason is not the correct explanation of assertion.
 (3) Assertion is correct but reason is incorrect.
 (4) Both assertion and reason are incorrect.

91. Match the column-I with Column-II:

	Column I		Column II
(A)	Unit of rate of reaction	(p)	$\text{mol L}^{-1} \text{s}^{-1}$
(B)	Unit of rate constant of zero order reaction	(q)	atm s^{-1}
(C)	Unit of rate constant of first order reaction	(r)	$\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
(D)	Unit of rate constant of third order reaction	(s)	s^{-1}

- (1) (A) $\rightarrow$ (p,q); (B) $\rightarrow$ (p,q); (C) $\rightarrow$ (s); (D) $\rightarrow$ (r)
 (2) (A) $\rightarrow$ (p); (B) $\rightarrow$ (r); (C) $\rightarrow$ (s); (D) $\rightarrow$ (q)
 (3) (A) $\rightarrow$ (p); (B) $\rightarrow$ (q); (C) $\rightarrow$ (r); (D) $\rightarrow$ (s)
 (4) (A) $\rightarrow$ (r); (B) $\rightarrow$ (q); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q)

92. Calculate freezing point of an aqueous solution of 5% MgCl_2 by weight if MgCl_2 shows 90% ionization ? [$K_f = 1.86 \text{ K.kg mol}^{-1}$]

- (1) -2.884°C (2) -1.030°C
 (3) -4°C (4) -3.2°C

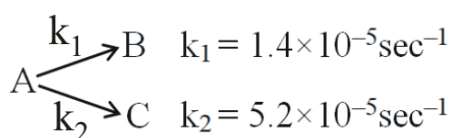
93. Which of the following is the feature of positive deviated solution ?

- (1) Solute-solute interaction < Solute-Solvent interaction
 (2) Solute-solute interaction > Solute-Solvent interaction
 (3) Solute-solute interaction = Solute-Solvent interaction
 (4) All of these

94. The electrical resistance of a column of 0.04 mol/L $\text{Ba}(\text{OH})_2$ solution of diameter 2 cm and length 100 cm is 4.44×10^3 ohm then equivalent conductivity of solution is :

- (1) $177.5 \text{ S cm}^2 \text{ eq}^{-1}$
- (2) $44.25 \text{ S cm}^2 \text{ eq}^{-1}$
- (3) $88.75 \text{ S cm}^2 \text{ eq}^{-1}$
- (4) $8.875 \text{ S cm}^2 \text{ eq}^{-1}$

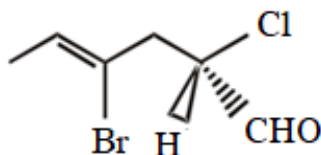
95.



The percentage distribution of B and C are :

- (1) 75% (B) & 25% (C)
- (2) 60% (B) & 40% (C)
- (3) 21.2 % (B) & 78.8 % (C)
- (4) 65.24% (B) & 34.76 % (C)

96. The IUPAC name of given compound is :-



- (1) (2R, 4E)-4-bromo-2-chloro-2-methylhexenal
- (2) (2S, 4Z)-4-bromo-2-chloro-4-hexenal
- (3) (2Z, 5R)-3-bromo-5-chloro-5-formyl-2-hexene
- (4) (2S, 4E)-4-bromo-2-chloro-2-methyl-4-hexenal

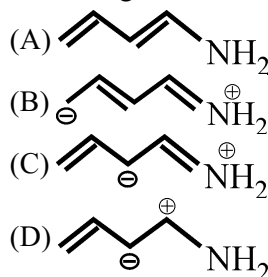
97. Which of the following structure has the R configuration at the chiral centre :-

- (1) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{COOH} \end{array}$
- (2) $\begin{array}{c} \text{H} \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2\text{CH}_3 \\ | \\ \text{Br} \end{array}$
- (3) $\begin{array}{c} \text{OH} \\ | \\ \text{CH}_3 - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{CH}_3 \end{array}$
- (4) $\begin{array}{c} \text{OH} \\ | \\ \text{H} - \text{C} - \text{CH}_3 \\ | \\ \text{CH}_2\text{CH}_3 \end{array}$

98. Select incorrect statement about enantiomers-

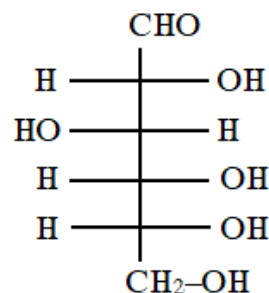
- (1) Equimolar mixture of enantiomers is optically inactive.
- (2) Enantiomers are optically active isomers.
- (3) Enantiomers are super imposable mirror images.
- (4) All are incorrect.

99. Arrange in decreasing order of stability to given resonating structures.



- (1) $A > B > D > C$
- (2) $A > B > C > D$
- (3) $A > C > B > D$
- (4) $B > A > C > D$

100. If given is D-glucose then identify L-glucose from options-



D-glucose

- (1) $\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{-OH} \end{array}$
- (2) $\begin{array}{c} \text{CHO} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{-OH} \end{array}$
- (3) $\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{-OH} \end{array}$
- (4) $\begin{array}{c} \text{CHO} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{-OH} \end{array}$

Topic : HUMAN GENOME PROJECT, DNA FINGERPRINTING, BIOTECHNOLOGY-I & II (COMPLETE), HUMAN HEALTH AND DISEASE, ECOLOGY [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-I)

101. In human :-

- I. Coding DNA is abundant.
- II. More than 2% of genome codes for protein.
- III. The function of more than 50% genes are unknown
- IV. Total number of genes is 30000.

Correct option are :-

- (1) III and IV (2) I and III
- (3) I, II and IV (4) I, II and III

102. 99.9% of base sequences among the humans are the same and the human genome has 3×10^9 bp. Find out, how many base sequences would be the different in two individuals.

- (1) 3×10^5 bps
- (2) 3×10^6 bps
- (3) 3×10^7 bps
- (4) 3×10^8 bps

103. A DNA sequence is polymorphic if -

- (1) The carrier frequency is less than 2 percent
- (2) It can be cut with a restriction enzyme
- (3) A variant sequence occurs in more than 1 percent of the population
- (4) It is present in more than 100 organisms

104. Which of the following is not a part of biotechnology:-

- (1) Gene therapy
- (2) Invitro fertilization leading to a test tube baby
- (3) Developing of DNA vaccine
- (4) Biomagnification of pesticide

105. Two enzymes responsible for restricting the growth of bacteriophages in *E. coli* were isolated. One was methylase and other was restriction endonuclease.

What is the significance of methylase ?

- (1) Protection of host DNA from the action of restriction endonuclease by adding methyl group to one or two bases usually within the sequence recognised by restriction enzyme
- (2) Able to ligate the two cohesive ends of DNA molecule
- (3) Able to remove the methyl group and hence, prevent the action of restriction endonuclease on host DNA
- (4) Able to cut the DNA of bacteriophage at specific sites

106. When foreign DNA is inserted into any vector, it results in inactivation of any marker gene. This is used for selection of :-

- (1) Non-Transformant cells
- (2) Transformant cells
- (3) Recombinant cells
- (4) (1) & (2) both

107. Which of the following statement is correct in the context of observing DNA fragments separated by agarose gel electrophoresis ?

- (1) DNA can be seen in visible light
- (2) DNA can be seen without staining in visible light
- (3) Ethidium bromide stained DNA can be seen in visible light
- (4) Ethidium bromide stained DNA can be seen under exposure to UV light

108. Restriction enzymes which cut the DNA a little away from the centre of palindrome site will produce :-

- (1) Sticky ends
- (2) Blunt ends
- (3) Either sticky or blunt
- (4) Unequal fragments

109. Which of the following is not correctly described.

- (1) **Transformation** :- is a process through which a piece of DNA is introduced in a host bacterium
- (2) **Elution** :- in this step, the separated bands of DNA are cut out from the agarose gel and extracted from the gel piece
- (3) **Micro-injection** :- is a vector mediated gene transfer best for animal cell
- (4) **PCR** :- In this multiple copies of the interest gene is synthesised in vitro

110. In agarose gel electrophoresis :

- (1) DNA migrates towards negative electrode.
- (2) Larger molecules migrate faster than smaller molecules
- (3) Smaller molecule migrate faster than larger molecule
- (4) Pink coloured bands of DNA can be visualised when gel is exposed to UV light.

111. **Assertion** : Unless one cuts the vector and the source DNA with the same restriction enzyme, the recombinant vector molecule cannot be created.

Reason : When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky ends'.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

112. **Statement-I** : Stirred Tank Reactor are normally Cylindrical .

Statement-II : The stirrer facilitates even mixing and oxygen availability throughout the bioreactor.

- (1) Both Statement-I and Statement-II is incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

113. Match the columns :-

Column A		Column B	
(i)	Ori	(A)	Cloning vector
(ii)	Ampicillin resistant gene	(B)	Cut DNA at restriction site
(iii)	Ti-plasmid	(C)	Selectable marker
(iv)	Restriction enzyme	(D)	Sequence from where replication starts

- (1) i-C, ii-D, iii-A, iv-B
- (2) i-D, ii-C, iii-B, iv-A
- (3) i-D, ii-C, iii-A, iv-B
- (4) i-C, ii-D, iii-B, iv-A

114. Select the correct match :

(1)	Insertional inactivation	Differentiate recombinants from non-recombinants
(2)	Recombinant protein	Any protein encoding gene is expressed in a homologous host
(3)	Down-stream processing	A series of process which start from raw material and going on up to completion of biosynthetic product.
(4)	Cloning	Introduction of the identified DNA into the host.

115. Select the correct match with respect to findings of HGP.
- (1) Maximum number of genes - chromosome 1
 - (2) Smallest gene → X chromosome
 - (3) EST → 1.4 million locations
 - (4) Chromosome-1 → Largest gene.
116. Which of the following is a transgenic plant ?
- (1) *Arabidopsis thaliana*
 - (2) *Flavr-Savr*
 - (3) *Meloidogyne incognita*
 - (4) *Caenorhabditis elegans*
117. Which of the following statement is correct -
- (1) Restriction endonuclease enzymes are used in isolation of plasmid
 - (2) Cry1Ac endotoxins obtained by *Bacillus thuringiensis* are effective against cotton bollworm.
 - (3) First transgenic dog 'ANDI' produced milk which was human α -antitrypsin enriched
 - (4) If any protein encoding gene is expressed in a homologous host is called recombinant protein
118. Which type of vectors were used for introducing nematode-specific genes in the host plant :-
- (1) Plasmid vectors
 - (2) Retrovirus
 - (3) Phagemid vectors
 - (4) *Agrobacterium* vectors
119. The cry II Ab and cry I Ab produce toxins that control :-
- (1) Cotton bollworms and corn borers respectively
 - (2) Corn borers and cotton bollworms respectively
 - (3) Tobacco budworms and corn borers respectively
 - (4) Corn borers and tobacco budworms respectively
120. ELISA is :-
- (1) Technique of disease diagnosis based on PCR
 - (2) Technique of gene therapy based on antigen-antibody interaction.
 - (3) Technique of molecular diagnosis based antigen-antibody interaction
 - (4) Technique of conventional diagnosis
121. From all transgenic animals, 95 percent of all existing transgenic animals are :-
- (1) pig (2) sheep (3) cow (4) mice
122. Which of the following could be permanent treatment of ADA deficiency ?
- (1) ADA enzyme replacement therapy
 - (2) Bone marrow transplantation
 - (3) The correct gene isolated from bone marrow cells producing ADA is introduced into cells at early embryonic stage
 - (4) Introduction of correct ADA gene at childhood stage
123. Select a *Coleoptera* insect that killed by action of Bt toxin :-
- (1) Beetles (2) Flies
 - (3) Army worm (4) Tobacco budworm
124. The trigger for activation of toxin of *Bacillus thuringiensis* is :-
- (1) High temperature
 - (2) Acidic pH of stomach
 - (3) Alkaline pH of gut of insect
 - (4) Mechanical action in the insect gut.
125. Ecological niche of an organism represents.
- a. The resources it utilizes
 - b. The range of conditions that it can tolerate
 - c. Its functional role in the ecological system
- (1) Only a (2) Only b and c
 - (3) Only a and c (4) All a, b and c

126. Which of the following organism breeds only once in their life time?
- Pacific salmon fish
 - Oyster
 - Pelagic fishes
 - Snails
127. If N is the population density at time t , then its density at time $t + 1$ is :-
- $N_{t+1} = N_t + [(B + E) - (D + I)]$
 - $N_{t+1} = N_t + [(B + D) - (E + I)]$
 - $N_{t+1} = N_t - [(B + I) - (D + E)]$
 - $N_{t+1} = N_t + [(B + I) - (D + E)]$
128. Given below are two statements:
- Statement-I** : Emigration is the number of individuals of same species who left the habitat and gone elsewhere during the time period under consideration.
- Statement-II** : Natality and Immigration contribute to an increase in population density.
- In the light of the above statements, choose the most appropriate answer from the options given below:-
- Both Statement-I and Statement-II are incorrect.
 - Statement-I is correct but Statement-II is incorrect.
 - Statement-I is incorrect but Statement-II is correct.
 - Both Statement-I and Statement-II are correct.
129. In Verhulst pearl logistic growth
- $$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right) \text{ Here, } \left(\frac{K - N}{K} \right) \text{ is :-}$$
- Carrying capacity
 - Intrinsic rate of natural increase
 - Environmental resistance
 - Biotic potential and environmental resistance
130. Ecology at the organismic level is essentially _____ ecology.
- Population
 - Ecosystem
 - Physiological
 - All of the above
131. **Statement-I:** Ecology is basically concerned with three levels of biological organisation-organisms, populations and communities.
- Statement-II:** On planet earth life only exist in a few favourable habitats not in extreme and harsh habitats.
- Both Statement I and Statement II are incorrect.
 - Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct
132. The Government of India established the National committee for environmental planning and coordination in :-
- 1984
 - 1972
 - 1872
 - 1884
133. Which of the following highest level in ecological Hierarchy consist of only biotic factors ?
- Population
 - Community
 - Biome
 - Biosphere
134. The overlapping area between two ecosystem where diversity is maximum is known as
- Community
 - Ecosystem
 - Ecotone
 - Habitat

135. Key stone species deserves protection because these :-

- (1) Have become rare due to over exploitation
- (2) Establishes essential connection between two components of nature
- (3) Are capable of surviving in harsh environment
- (4) Play an important role in supporting the species in community

SECTION-B (BIOLOGY-I)

136. Arrange the following in correct order of sequence for DNA finger-printing-

- (a) Blotting
- (b) Hybridization
- (c) Gel Electrophoresis
- (d) Autoradiography

- (1) c, a, d, b (2) a, c, d, b
- (3) b, c, a, d (4) c, a, b, d

137. The commonly used vectors in HGP are :-

- (1) Plasmid (2) Cosmid
- (3) BAC & YAC (4) Phasmid

138. BamH I site is present at :-

- (1) amp^R (2) tet^R
- (3) *Ori* (4) *rop*

139. Which of the DNA sequence could be possible recognition sequence of a restriction endonuclease?

- (1) 5' G C G C G C 3'
3' C G C G C G 5'
- (2) 5' G T T C A T 3'
3' C A A G T A 5'
- (3) 5' C A A T G 3'
3' G T T A C 5'
- (4) 5' G C A T T A 3'
3' C G T A A T 5'

140. **Assertion** : Origin of replication (*ori*) is responsible for controlling the copy number of the linked DNA.

Reason : Any piece of DNA when linked to origin of replication can be made to replicate within the host cells.

- (1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
- (2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
- (3) Assertion is true but the reason is false.
- (4) Both assertion & reason are false.

141. **Statement-I** : Biolistic method is used for gene transfer in plant.

Statement-II : DNA is coated on Gold / Tungsten Particle in Gene Gun method.

- (1) Both Statement-I and Statement-II is incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

142. Transgenic plants are the ones :-

- (1) Grown in artificial medium after hybridization in the field
- (2) Produced by a somatic embryo in artificial medium
- (3) Generated by introducing foreign DNA in to a cell and regenerating a plant from that cell
- (4) Produced after protoplast fusion in artificial medium

143. Read the following statements :-

- (a) Nematode resistant plant developed by novel strategy based on process of RNA interference (RNAi).
- (b) RNAi takes place in all eukaryotic organisms as method of cellular defence.
- (c) RNAi involves silencing of specific mRNA due to complementary dsRNA.
- (d) Transgenic host expressing two non-complementary RNA's that initiated RNAi.

Correct statements are :

- (1) Only a,b (2) Only a,b,d
- (3) Only a, b, c (4) a, b, c, d

144. It present, about how many recombinant therapeutics have been approved for human use in India.

- (1) 30 (2) 12
- (3) 1000 (4) 200,000

145. Which human protein is used to treat emphysema ?

- (1) α -1-antitrypsin (2) α -1-lactalbumin
- (3) α -amylase (4) β -galactosidase

146. **Assertion :-** Population ecology is an important area of ecology.

Reason :- Evolutionary changes through natural selection takes place at population level.

In the light of above statements choose the most appropriate answer from the options given below.

- (1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
- (2) Assertion is true but Reason is false
- (3) Assertion is false but Reason is true
- (4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

147. In a pond there were 40 lotus plants last year and through reproduction 10 new plants are added, taking the current population to 50, then we calculate the birth rate as

- (1) 0.4 offspring per lotus per year
- (2) 0.52 offspring per lotus per year
- (3) 0.25 offspring per lotus per year
- (4) 0.2 offspring per lotus per year

148. Recognise the table and find out the correct statement.

	Species A	Species B
(a)	+	+
(b)	-	-
(c)	+	-
(d)	+	-
(e)	+	0
(f)	-	0

- (1) a-competition, b-predation, c-commensalism, d-parasitism, e-amensalism, f-mutualism
- (2) b-competition, c-predation, f-commensalism, d-parasitism, e-amensalism, a-mutualism
- (3) b-competition, c-predation, e-commensalism, d-parasitism, f-amensalism, a-mutualism
- (4) b-competition, d-predation, c-commensalism, e-mutualism, f-amensalism, a-parasitism

149. Verhulst-Pearl logistic growth pattern is

- (1) Sigmoid (2) J-shaped
- (3) Straight line (4) Hyperbola

150. Which of the following is not an attribute of population?

- (1) Biotic potential (2) Dominance
- (3) Natality (4) Age pyramid

Join Telegram: @NEETxNOAH

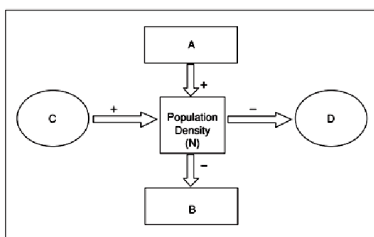
Topic : HUMAN GENOME PROJECT, DNA FINGERPRINTING, BIOTECHNOLOGY-I & II (COMPLETE), HUMAN HEALTH AND DISEASE, ECOLOGY [**BACK UNIT**] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-II)

151. A species of mammal is found only in a certain part of eastern Himalayas and is not found anywhere else. Which of the following term will be used for such species?

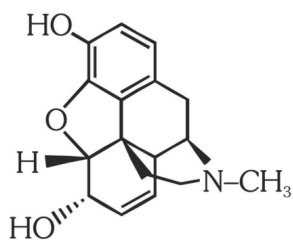
- (1) Keystone species (2) Critical link species
(3) Alien species (4) Endemic species

152. Identify A, B, C and D in this figure.



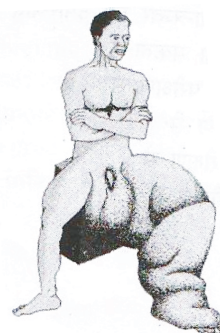
- (1) A: Immigration (I), B: Emigration (E), C: Natality (B), D: Mortality (D)
- (2) A: Natality (B), B: Immigration (I), C: Mortality (D), D: Emigration (E)
- (3) A: Mortality (D), B: Emigration (E), C: Natality (B), D: Immigration (I)
- (4) A: Mortality (D), B: Natality (B), C: Emigration (E), D: Immigration (I)

153. Which of the following statement is true for the structure given below :-



- (1) It interferes with the transport of the neurotransmitter dopamine
- (2) It is useful in patients who have undergone surgery
- (3) Obtained from *Cannabis Sativa*
- (4) It has a potent stimulating action on central nervous system

154. Read the following statements (a–e) :



- (a) Gross deformities in genital organs
- (b) Elephantiasis
- (c) Vector is male *Aedes* mosquito
- (d) Slowly developing chronic inflammation, usually of blood vessels of lower limbs
- (e) It is caused by *Wuchereria bancrofti*.

How many of the above statements are incorrect about given diagram?

- (1) One (2) Two
(3) Three (4) Four

155. Observe the following table very carefully contain source of drug, biological name and their uses. Choose the correct option :

(1)		<i>Papaver somniferum</i>	Stimulant action on body
(2)		<i>Claviceps purpurea</i>	Hallucination
(3)		<i>Papaver somniferum</i>	Depressant action on body
(4)		<i>Cannabis sativa</i>	Relieving pain

156. Read the following statements (A – D).

(A) *Haemophilus influenzae* is responsible for the disease pneumonia in human.

(B) *Rhino* Viruses cause one of the most infectious human ailments common cold.

(C) *Wuchereria* cause filariasis.

(D) Many fungi like *Microsporum*, *Trichophyton* and *Epidermophyton* are responsible for ringworms.

How many of the above statement are true :-

(1) 4 (2) 3 (3) 2 (4) 1

157. Select the set of correct statements from the options given below.

(A) Acquired immunity is pathogen specific.

(B) Neutrophils and macrophages form cellular barrier of innate immunity.

(C) Humoral immunity is provided by B - cells

(D) Each antibody molecule has four peptide chain.

(1) A & B are correct

(2) B & C are correct

(3) C & D are correct

(4) A, B, C & D are correct

158. Given below are two statements:

Statement I : Techniques like radiography and computed tomography uses X-rays to detect cancers.

Statement II : MRI uses strong magnetic fields and ionising radiations to detect pathological and physiological changes in the living tissue.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

159. Which one of the following options gives the correct matching of a disease with its causative organism and mode of infection?

	Disease	Causative Organisms	Mode of Infection
(1)	Elephantiasis	<i>Wuchereria bancrofti</i>	With infected water and food
(2)	Malaria	<i>Plasmodium vivax</i>	Bite of male Anopheles mosquito
(3)	Typhoid	<i>Salmonella typhi</i>	With inspired air
(4)	Pneumonia	<i>Streptococcus pneumoniae</i>	Droplet infection

160. Read the following statements and select the correct statement :-

(A) HIV infects and destroys the helper T-cells.

(B) HIV is a retrovirus.

(C) HIV infection is the most dangerous STD as it is non-curable.

(D) The newer antiretroviral drugs are a cure for HIV infection.

(E) HIV contains dsRNA & some enzymes like reverse transcriptase, protease and integrase.

(1) A,C,E

(2) A,B,C

(3) A,B,C,D

(4) B,C,D & E

161. Given below are two statements :

Statement I : In the primary lymphoid organs immature lymphocyte differentiate into antigen-sensitive lymphocytes.

Statement II : The secondary lymphoid organ provide the site for interaction of lymphocytes with the antigen and also for proliferation.

(1) Both statement I and statement II are false

(2) Both statement I and statement II are true

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

162. How many statements are correct for typhoid?

- (i) It is a bacterial disease
- (ii) Pathogen enters small intestine through contaminated food and water.
- (iii) Sustained Low fever (39-40°F)
- (iv) Typhoid Fever could be confirmed by widal test

(1) Three (2) Four (3) Two (4) One

163. What happens when cells sloughed from malignant tumors reach distant sites in the body?

- (1) They become dormant
- (2) They undergo apoptosis
- (3) They start a new tumor
- (4) They lose their ability to divide

164. What role do interferons play in innate immunity?

- (1) They stimulate antibody production
- (2) They directly kill pathogens
- (3) They prevent non-infected cells from viral infection
- (4) They enhance phagocytosis by macrophages

165. How do vaccines help in neutralizing pathogenic agents during actual infection?

- (1) By directly attacking the pathogens
- (2) By generating memory B and T-cells
- (3) By decreasing production of cytokines
- (4) By inducing inflammation

166. **Assertion** : Smoking causes oxygen deficiency in the body.

Reason : Smoking increases carbon monoxide (CO) content in blood and reduces concentration of haem bound oxygen.

- (1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
- (2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (3) **Assertion** is true but **Reason** is false.
- (4) **Assertion** is false but **Reason** is true.

167. **Assertion** :- Treatment of AIDS with anti-retroviral drugs is only partially effective.

Reason :- HIV infection, more often, spreads due to conscious behaviour pattern.

Choose the correct answer from the given options :-

- (1) Both Assertion & Reason are correct but Reason is not the correct explanation of the Assertion.
- (2) Assertion is correct but the Reason is not correct.
- (3) Both Assertion & Reason are incorrect.
- (4) Both Assertion & Reason are correct & the Reason is a correct explanation of the Assertion.

168. **Statement-I** :- Amniocentesis is legal under some conditions.

Statement-II :- Amniocentesis is legal for sex determination of foetus.

- (1) Statement-I & II both are correct
- (2) Statement-I & II both are incorrect
- (3) Statement-I is correct but II is incorrect
- (4) Statement-I is incorrect but II is correct

169. In uterus, endometrium proliferates in response to

- (1) Relaxin (2) Oxytocin
- (3) Progesterone (4) Oestrogen

170. Identify the hormones that are secreted in large amount prior to ovulation :

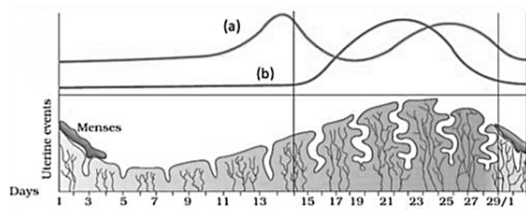
- A. LH B. FSH
- C. Estrogen D. Progesterone

- (1) A only (2) A & B only
- (3) A, B & C only (4) A, B, C & D

171. The mammary glands are paired structures that contain the glandular tissue and variable amount of fat. The correct sequence of tissues involved in synthesis and flow of milk are

- (1) Mammary lobes → Mammary alveoli → Mammary ampulla → Mammary duct → Lactiferous duct
- (2) Mammary lobes → Mammary alveoli → Mammary duct → Mammary ampulla → Lactiferous duct
- (3) Mammary lobes → Mammary alveoli → Lactiferous duct → Mammary ampulla → Mammary duct
- (4) Mammary alveoli → Mammary lobes → Lactiferous duct → Mammary duct

172. Which hormone is represented by (a) ?



- (1) LH (2) FSH
(3) Estrogen (4) Progesterone

173. ___A___ and uterine tissue become interdigitated with each other and jointly form placenta. Identify "A" here ?

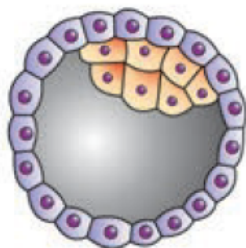
- (1) Amnion (2) Umbilical cord
(3) Chorionic villi (4) Trophoblast

174. **Assertion :** The cleavage starts as the zygote moves through the isthmus of the oviduct towards ampulla.

Reason : The embedment of the blastocyst in the endometrium of the uterus is called implantation.

- (1) Assertion is true but Reason is false.
(2) Assertion is false but Reason is true.
(3) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(4) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

175.



Identify the given figure and its site of embedment.

	Figure	Site
(1)	Zygote	Fallopian tube (Ampulla)
(2)	Morula	Fallopian tube (Isthmus)
(3)	Blastocyst	Uterus (Perimetrium)
(4)	Blastocyst	Uterus (endometrium)

176. The blastomeres in the blastocyst are arranged into an outer layer called ___A___ and an inner group of cells attached to trophoblast called the ___B___. Here A and B are respectively ?

- (1) A - trophoblast, B - inner cell mass
(2) A - inner cell mass, B - chorionic villi
(3) A - placental villi, B - inner cell mass
(4) A - trophoblast, B - chorionic villi

177. Which of the following statement is not correct regarding parturition ?

- (1) Oxytocin is the main hormone responsible for parturition.
(2) Signal for parturition originate from fully developed foetus only.
(3) During parturition estrogen progesterone ratio increases.
(4) Parturition is induced by a complex neuro-endocrine mechanism.

178. What does foetal ejection reflex means ?

- (1) Release of oxytocin from maternal pituitary.
(2) Mild uterine contractions.
(3) Expulsion of the baby from mother's birth canal.
(4) Decrease in estrogen level in maternal blood.

179. Which hormone acts on the uterine muscles and causes stronger uterine contractions ?

- (1) Oxytocin (2) Relaxin
(3) Cortisol (4) Progesterone

180. **Assertion :** The second meiotic division of the secondary oocyte completes during fertilisation.

Reason : The second meiotic division is unequal and results in the formation of a second polar body and a diploid Ovum.

- (1) Assertion is false but Reason is true.
(2) Assertion is true but Reason is false.
(3) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

181. Which of the following event doesn't occur in case of pregnancy?

- (A) Decrease in level of progesterone.
(B) Degeneration of corpus luteum.
(C) Stoppage of menstrual cycle.
(D) Endometrial hyperplasia and myometrial *hypertrophy*.
(E) Increase in level of thyroxine in maternal blood.

- (1) A, B and D (2) B, D and E
(3) A and B (4) A, B and E

182. **Statement-I** : The trophoblast layer gets attached to the endometrium and the inner cell mass gets differentiated as the embryo.

Statement-II : The morula becomes embedded in the endometrium of the uterus.

- (1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

183. Select the hormone which is not correctly matched with its function.

(A)	Estrogen – Endometrial hyperplasia
(B)	hCG – Maintains corpus luteum
(C)	Oxytocin – Strong uterine contractions
(D)	Prolactin – Milk synthesis
(E)	Progesterone – promote foetal ejection reflex

- (1) A and E (2) Only D
(3) D and E (4) Only E

184. Match column-I with column-II ?

Column-I		Column-II	
(a)	Gametogenesis	(i)	Delivery of the baby
(b)	Insemination	(ii)	Formation of gametes
(c)	Parturition	(iii)	Fusion of male & female gamete
(d)	Fertilisation	(iv)	Transfer of sperm into female tract genital
(e)	Implantation	(v)	Attachment of blastocyst to uterine wall

- (1) a–ii, b–i, c–iv, d–iii, e–v
(2) a–iii, b–i, c–iv, d–ii, e–v
(3) a–i, b–iii, c–v, d–ii, e–iv
(4) a–ii, b–iv, c–i, d–iii, e–v

185. Which of the following statement is not correct regarding testes ?

- (A) Testes are formed inside scrotum during embryonic life.
(B) Each testis has about 250 compartments called testicular lobules.
(C) Each testis contains one to three seminiferous tubules.
(D) Interstitial cells synthesise and secrete testicular hormones called ABP.

- (1) Only A
(2) Only C
(3) A, C, and D
(4) A and C

SECTION-B (BIOLOGY-II)

186. Recently chikungunya and dengue cases were reported from various parts of the country. Choose the correct vector responsible for these diseases

- (1) *Anopheles* mosquito
(2) *Xenopsylla* (Rat flea)
(3) *Culex* mosquito
(4) *Aedes* mosquito

187. Match the following columns and choose the **correct** answer from the options given below

	Column I		Column II
A	Physical barrier	i	PMNL
B	Physiological barrier	ii	Interferons
C	Cellular barrier	iii	Acid in stomach
D	Cytokine barrier	iv	Mucus coating of epithelial lining

- (1) A - iv, B - iii, C - i, D - ii
(2) A - iv, B - i, C - ii, D - iii
(3) A - iii, B - ii, C - iv, D - i
(4) A - ii, B - iv, C - iii, D - i

188. Match the column-I and column-II

Column-I		Column-II	
(a)	Primary lymphoid organs	(i)	Spleen
(b)	Secondary lymphoid organ	(ii)	Thymus
(c)	Bone Marrow	(iii)	Large bean shaped organ
(d)	Spleen	(iv)	Produced lymphocytes

- (1) a-(ii), b-(i), c-(iv), d-(iii)
 (2) a-(ii), b-(i), c-(iii), d-(iv)
 (3) a-(i), b-(ii), c-(iv), d-(iii)
 (4) a-(i), b-(ii), c-(iii), d-(iv)

189. Which of the following disease does not belong to the same category?

- (1) Typhoid (2) Plague
 (3) Diphtheria (4) Amoebiasis

190. How does heroin typically affect body functions?

- (1) By accelerating heart rate and respiration
 (2) By inducing hyperactivity and agitation
 (3) By acting as a stimulant on the central nervous system
 (4) By depressing body functions and slowing them down

191. **Statement I** : Injecting microbes during immunization induces passive immunity.

Statement II : AMI is responsible for graft rejection.

- (1) Both statement I and II are true
 (2) Only statement I is true
 (3) Only statement II is true
 (4) Both statement I and II are false

192. **Statement-I** : AIDS is a congenital disease.

Statement-II : AIDS is transmitted from mother to foetus.

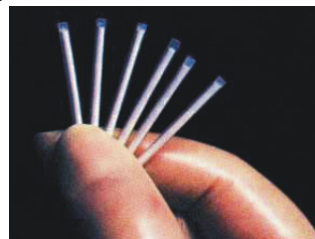
- (1) Both Statement I and Statement II are false.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are correct.

193. **Assertion** : Both bone marrow and thymus provide sites for interaction of lymphocytes with antigens.

Reason : After maturation the lymphocytes migrate to secondary lymphoid organ.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
 (2) **Assertion** is true but **Reason** is false.
 (3) **Assertion** is false but **Reason** is true.
 (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

194. Identify the contraceptive device shown below as well as the related right place of its implantation into a woman and select the correct option for the two together :



	Contraceptive device	Site of implant
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Implants	Subcutaneous
(4)	Multiload 375	Uterine wall

195. Match the column A with column B :

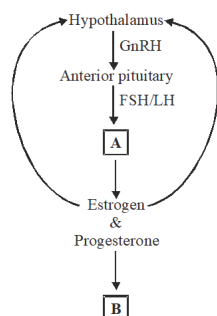
	Column-A		Column-B
A	Barrier method	1	Tubectomy
B	Oral pills	2	Condoms
C	IUD's	3	Prevents ovulation
D	Sterilisation	4	CuT

- (1) A-4 B-2 C-3 D-1
 (2) A-2 B-3 C-1 D-4
 (3) A-2 B-3 C-4 D-1
 (4) A-1 B-2 C-4 D-3

196. Which of the following method of birth control can cause excessive menstrual bleeding pain and inflammation of uterine wall ?

- (1) Condom (2) Cervical cap
 (3) Oral contraceptive (4) Intra uterine device

197.



Find out 'A' and 'B' respectively :-

- (1) Ovary and Uterus
- (2) Uterus and Ovary
- (3) Oviduct and Ovary
- (4) Seminal vesicle and Uterus

198. Which of the following cells have the potency to give rise to all the tissues and organs during embryonic development?

- (1) Trophoblast
- (2) Inner cell mass
- (3) Chorionic villi
- (4) Placental villi

199. Match column-I with column-II and select the correct option using the codes given below-

Column-I		Column-II	
(a)	Corpus luteum	(i)	Germinal layers
(b)	Clitoris	(ii)	Sperm
(c)	Nebenkern sheath	(iii)	Female external genitalia
(d)	Inner cell mass	(iv)	Endocrine structure

- (1) a-iv, b-iii, c-ii, d-i
- (2) a-i, b-iii, c-ii, d-iv
- (3) a-i, b-ii, c-iii, d-iv
- (4) a-iv, b-ii, c-iii, d-i

200. Identify the primary sex organ of male, its main hormone and its chemical nature respectively.

- (1) Testes, testosterone and glycoprotein
- (2) Testes, testosterone and steroidal
- (3) Ovary, estrogen and steroidal
- (4) Ovary, progesterone and glycoprotein

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